Priyanka Vijaybaskar

Pittsburgh, PA

412-224-8538 | pvijayba@andrew.cmu.edu | www.linkedin.com/in/priyankavijaybaskar

https://github.com/prinks-wiz | https://prinks-wiz.github.io/Priyanka-Vijaybaskar/

EDUCATION

Carnegie Mellon University — **Aug 2025 - Dec 2026**

Master's, Computational Data Science (GPA: 3.85/4.00) — *Pittsburgh, PA*

- **Coursework:** Cloud Computing, Neuro Symbolic AI

Sri Sivasubramaniya Nadar College of Engineering, Anna University — **Sep 2021 - Jun 2025**

Bachelors of Technology, Information Technology (GPA: 8.12/10) — *Chennai, India*

- **Achievements:** President, ProCoDe.

SKILLS

- **Frameworks and Libraries:** Tensorflow, Numpy, Keras, Pandas, Seaborn, Matplotlib, Langchain, LangGraph, Flask, PyTorch, FastAPI, Docker, Vertex AI, Azure, GCP, AWS, CI/CD, RAG, GraphRAG, OpenCV, AWS, Kubernetes, Maven, Software Engineering
- **Programming Languages:** Python, SQL, Java, JavaScript
- **Tools:** Git, NVIDIA Jetson, Weaviate, FAISS, W&B, TensorBoard, Azure, GCP, CPU Architecture

EXPERIENCE

Carnegie Mellon University – Space Research Argus | *Graduate Researcher* — **Oct 2025 - Present**

- Developed an end-to-end computer-vision pipeline for a **NASA-class CubeSat** mission, fine-tuning **EfficientNet** with PyTorch on **100k+ Sentinel-2 images** to classify over **15 geographic regions**, delivering reliable high-fidelity classifications that support mission planning
- Architected a **700x speedup** through custom GPU kernel optimizations on the **NVIDIA Jetson Orin**. Achieved 65% model compression and 40% latency reduction via **INT8 quantization** and **structured pruning** under strict 5W power constraints.
- Engineered high-fidelity simulation environments with Google Earth Engine data to test ML pipelines on unseen out-of-distribution geospatial inputs, confirming reliable performance across mission scenarios and reducing validation errors by 30%

Siam Computing | *Product Strategy Intern* — **Jun 2024 - Aug 2024**

- Authored 5+ comprehensive product documents including PRDs for **25+ features across HRM systems**, competitive analysis, FRDs, PRDs, defined success metrics and KPIs achieving **2.4-minute improvement in booking time.**
- Conducted user research with 25+ healthcare providers and analyzed **1,000+ user interactions using Mixpanel,** which informed product decisions and improved workflow efficiency; managed stakeholders across medical staff, engineering, and C-suite leadership

Finsire Technologies | *Data Science Intern* — **Jan 2024 - Apr 2024**

- Engineered **time-series forecasting** model using Continuous Wavelet Transform for feature engineering on **500K+ mutual fund records** from MFI API; implemented hyperparameter optimization improving prediction **accuracy 15%** over baseline LSTM, directly securing 3 institutional clients.
- Built automated ETL pipelines for data extraction of **2K real estate listings** and SEBI financial statements; developed TF-IDF semantic matching algorithm with processing 50K+ vehicle-insurance pairs achieving **92% precision** for deployment.

PROJECTS

Cloud-Native Twitter Recommendation Service on AWS — **Jan 2026 - Present**

Carnegie Mellon University

- Designed and deployed a Kubernetes-based microservice architecture on AWS using kOps, serving a Twitter recommendation engine at 8,700+ RPS with 93.5% correctness across a heterogeneous cluster of ARM-based Graviton instances.
- Achieved a 41.7x throughput improvement through systematic bottleneck elimination: schema denormalization (3.5x), InnoDB buffer pool tuning to 98.4% cache hit ratio, prepared statement caching, and a 45-second auth token TTL that reduced authentication overhead by 95%.
- Built a fully automated CI/CD pipeline with GitHub Actions and Helm, supporting immutable SHA-tagged deployments, conditional database initialization, and NLB-based traffic routing, enabling cluster recreation in under 8 minutes.

Data Influence for Vision-Language Models (VLM) — **Feb 2026 - Present**

Carnegie Mellon University

- Fine-tuned VLMs via LoRA (training <2% of 7B parameters) with 7 A100-targeted optimizations including gradient checkpointing, TF32 precision, and 8-bit AdamW, reducing training time from days to hours on a single 40GB GPU.
- Diagnosed a label-source mismatch in autonomous driving action prediction where GPS-derived waypoint labels conflict with visual scene content, then built a VQA pipeline integrating navigation context with LLaVA-1.5-7B and Qwen2.5-VL to achieve 3.5x accuracy over vision-only baselines across 16K+ driving samples.

Physics-Based Motion Prediction using Neural ODE-VAE Architecture — **Sep 2025 - Dec 2025**

Carnegie Mellon University

- Benchmarked multiple generative models (GANs, Pix2pix, diffusion models, VLM fine-tuning) for **trajectory prediction** on PhysicsGen dataset; designed novel Neural ODE-VAE hybrid architecture improving upon **baseline by 75%**.
- Implemented dynamics modelling with Neural ODEs integrated into VAE latent space, enabling physically-consistent trajectory generation of images.